RESEARCH SEMINAR • APPLIED MATHEMATICAL FINANCE

Financial Decision Support System (FDSS)

A Mathematical & Stochastic Framework for Retail Financial Risk Modeling

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Research Category: Category 3: Pure Sciences (Mathematics & Statistics)
Focus: Numerical Root-Finding & Stochastic Calculus (GBM)
Repository: github.com/AlmightyXZen/Financial-Decision-Support-System

1. The Research Problem: The Deterministic Fallacy

Why conventional retail financial calculators are mathematically flawed and mislead the public

The Deterministic Expectation Assumption

- Commercial calculators project future wealth using single-point linear formulas: $A = P(1 + r)^t$.
- They assume an invariant constant rate of return (e.g., exactly 12.00% every single year for 20 years).
- **The Flaw:** Real capital markets do not follow deterministic lines; they are governed by stochastic Brownian motion with non-zero variance $\sigma^2 > 0$.

Theoretical Violation: Jensen's Inequality

For a convex function $g(x)$, $\blacksquare[g(X)] \neq g(\blacksquare[X])$. Due to volatility drag, compounding under uncertainty yields a median trajectory strictly below the arithmetic average.

Empirical Consequences for Citizens

- **Retirement Shortfall:** Investors base life savings on optimistic straight lines, experiencing catastrophic downside shortfalls.
- **Predatory Credit Traps:** Borrowers misunderstand compounding frequencies and fall into negative amortization loans.
- **Absence of Tail Risk:** Retail platforms completely omit Value at Risk (VaR) and goal attainment probabilities.

The Research Gap: Institutional quantitative hedge funds model full stochastic distributions, while everyday citizens are relegated to flawed, 18th-century deterministic formulas.

2. Research Hypothesis & Core Objectives

Transforming financial decision support from static scalar estimates into rigorous probability densities

Central Research Hypothesis

"By replacing deterministic point expectations with numerical root-finding algorithms and continuous stochastic processes (Geometric Brownian Motion), retail investors can be equipped with robust, distribution-aware decision support that quantifies downside tail risk and eliminates the volatility drag illusion."

Objective 1: First-Principles Theory

Derive all financial laws from axiomatic foundations: governing ODEs, Euler limits, finite geometric series, and stochastic calculus.

Objective 2: Numerical Optimization

Implement high-precision root-finding (Newton–Raphson & Bisection) to solve multi-period non-linear equations where Galois theory proves algebraic radicals fail.

Objective 3: Stochastic Risk Simulation

Simulate 10,000+ stochastic price trajectories to compute terminal wealth densities, empirical goal attainment odds, and 95% Value at Risk (VaR).

3. Theoretical Pillar 1: Time Value of Money & Governing ODEs

Comparing linear accumulation, discrete geometric progression, and continuous infinitesimal limits

Linear vs. Continuous Compounding ODEs

• **Simple Interest (Linear ODE):** Rate of growth is constant across all time:

$dA/dt = P \cdot r, \quad A(0) = P \Rightarrow \mathbf{A(t) = P(1 + rt)}$

Violates temporal subadditivity: $A(t_1 + t_2) < A(t_1)A(t_2)/P$.

• **Continuous Compounding (Homogeneous Linear ODE):** Rate of growth is proportional to instantaneous wealth:

$dA/dt = r \cdot A(t), \quad A(0) = P \Rightarrow \mathbf{A(t) = P\, e^{rt}}$

Euler's Limit as Compounding Frequency $m \to \infty$

$$\lim_{m\to\infty} P(1 + r/m)^{mt} = P\left[\lim_{x\to\infty} (1 + 1/x)^{x}\right]^{rt} = P\, e^{rt}$$

Discrete to Continuous Yield Equivalence

To match the terminal wealth of an annual rate r_d compounded m times with a continuous rate r_c:

$e^{r_c} = (1 + r_d/m)^m \Rightarrow \mathbf{r_c = m \cdot \ln(1 + r_d/m)}$

Effective Annual Rate (EAR / APY)

Exposing the retail 'APR illusion':

$\mathbf{EAR = (1 + r_{nom}/m)^m - 1} \quad • \quad \mathbf{EAR_{cont} = e^{r_{nom}} - 1}$

A credit card charging 18.00% APR compounded monthly produces a true annualized cost of **19.56% EAR**.

The Stochastic Discount Factor

Present Value operator: $\mathbf{PV = FV \cdot (1 + r/m)^{-mt}} \quad • \quad \mathbf{PV_{cont} = FV \cdot e^{-rt}}$

4. Theoretical Pillar 2: Consumer Debt & Negative Amortization

Finite geometric series summation and the mathematical boundary condition for debt traps

Analytical Derivation of Equated Monthly Installment (EMI)

Equating the borrowed Principal P to the discounted present value of an n-period annuity with monthly interest rate $R = r/12$:

$$P = \sum_{k=1}^{n} E \cdot (1 + R)^{-k} = E \cdot [\, (1 - (1 + R)^{-n}) / R \,]$$

Inverting the geometric sum yields the exact closed-form EMI equation:

$$\boldsymbol{E = P \cdot R \cdot (1 + R)^{n} / [\, (1 + R)^{n} - 1 \,]}$$

Principal / Interest Amortization Dynamics

At month k: Interest component $I_k = B_{k-1} \cdot R$, Principal repaid $P_k = E - I_k$.
Principal repayments form an escalating geometric progression: $P_k = P_1(1 + R)^{k-1}$.

The Negative Amortization Boundary Condition

In retail finance, predatory 'teaser rate' loans or minimum credit card payments often charge an installment E below the monthly accrued interest.

Boundary Theorem: A loan amortizes toward zero if and only if:

$$\boldsymbol{E > P \cdot R \iff E > P \cdot (r / 12)}$$

If $E \leq P \cdot R$, outstanding debt balance diverges: $\lim_{n \to \infty} B_n = +\infty$.

Societal Relevance for Consumer Protection

Our system implements automated boundary testing: detecting predatory credit structures and displaying exact inflection points where prepayments collapse total interest expense.

5. Theoretical Pillar 3: Numerical Optimization for Yield Analysis

The Abel–Ruffini impossibility theorem and high-order Newton–Raphson root finding

The Internal Rate of Return (IRR) Polynomial

For irregular cash flows $\{C_0, C_1, ..., C_T\}$, the IRR is the discount rate r^* setting Net Present Value to zero:
$\text{NPV}(r) = \sum_{t=0}^{T} C_t / (1 + r)^t = 0$

Galois Theory & The Abel–Ruffini Theorem

Setting $x = 1 / (1 + r)$ transforms $\text{NPV}(r) = 0$ into a polynomial of degree T: $\sum C_t x^t = 0$.
The Abel–Ruffini Theorem proves that general algebraic solutions in radicals **do not exist for degree $T > 4$**.
Therefore, calculating multi-year financial yields **strictly necessitates numerical analysis**.

Quadratic Convergence via Newton–Raphson

We utilize the analytical first derivative with respect to discount rate r:
$f'(r) = -\sum_{t=1}^{T} [\, t \cdot C_t / (1 + r)^{t+1} \,]$
Iterative update: $\mathbf{r_{k+1} = r_k - [\, NPV(r_k) / f'(r_k) \,]}$
Exhibits quadratic convergence: $|e_{k+1}| \le M |e_k|^2$ (error squares every step).

Robustness: Bisection Fallback Mechanism

When non-conventional cash flows exhibit multiple sign changes (Descartes' Rule of Signs) or $f'(r) \to 0$, our engine automatically falls back to interval bisection on $[a, b]$ ensuring guaranteed linear convergence.

6. Theoretical Pillar 4: Stochastic Asset Dynamics & Itô's Lemma

Continuous-time Brownian motion and the mathematical proof of the volatility drag gap

The Stochastic Differential Equation (GBM)

Asset prices evolve under continuous drift μ and volatility σ driven by a standard Wiener process (Brownian Motion) W_t:

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t \qquad (dW_t \sim \blacksquare(0, dt))$$

Proof of Volatility Drag via Itô's Lemma

In standard calculus, $d(\ln S) = dS/S$. But Brownian paths have non-zero quadratic variation: $(dW_t)^2 = dt$.
Applying Itô's Lemma to $f(S) = \ln S$ with $f' = 1/S$ and $f'' = -1/S^2$:

$$d(\ln S) = (1/S)dS + (1/2)(-1/S^2)(dS)^2$$
$$d(\ln S) = (\mu\, dt + \sigma\, dW) - (1/2)\sigma^2\, dt$$
$$\boldsymbol{d(\ln S) = (\mu - \tfrac{1}{2}\sigma^2)\, dt + \sigma\, dW_t}$$

The Exact Discrete Path Solution

Integrating yields the exact trajectory for time step Δt:

$$S_{t+\Delta t} = S_t \cdot \exp\left[(\mu - \tfrac{1}{2}\sigma^2)\Delta t + \sigma \sqrt{\Delta t} \cdot Z \right] \quad (Z \sim \blacksquare(0, 1))$$

The Fundamental Divergence: Mean vs. Median

- **Arithmetic Mean:** $\blacksquare[S_t] = S_0\, e^{\mu t}$ (pulled up by extreme right-tail skewness).
- **Typical Median Path:** $S_{med}(t) = S_0\, e^{(\mu - \frac{1}{2}\sigma^2)t}$.
- **Structural Volatility Drag:** $\Delta\mu = -\tfrac{1}{2}\sigma^2$.

Key Finding: For a stock fund with $\mu = 12\%$ and $\sigma = 20\%$, drag is –2.00% per year! Over 10 years, median wealth lags arithmetic expectation by **INR 601,835 per Million invested**. More than 50% of investors underperform the average!

7. Theoretical Pillar 5: Monte Carlo Simulation & Risk Estimation

Large-scale stochastic ensembles, Central Limit Theorem bounds, and downside tail risk metrics

Monte Carlo Ensemble & CLT Convergence

We generate M = 10,000+ independent market paths using vectorized NumPy operations.
By the **Strong Law of Large Numbers (SLLN)**, the sample average converges almost surely to the true expectation: $\lim (1/M)\sum g(S_T) = \mathbb{E}[g(S_T)]$.
By the **Central Limit Theorem**, standard error decays at the canonical rate:
Standard Error = $\sigma_{sample} / \sqrt{M} = \mathbb{E}(1 / \sqrt{M})$
For M = 10,000, estimator precision narrows to ±1% of sample variance.

Quantitative Downside Risk Metrics

- **Empirical Goal Attainment Probability:**
  $P\mathbb{E}(S_T \geq \text{Goal}) = (1 / M) \sum_{m=1}^{M} \mathbb{E}_{\{S_{T,m} \geq \text{Goal}\}}$
- **Value at Risk (VaR$_\alpha$):** Maximum expected loss at confidence level $(1 - \alpha)$ over horizon T:
  $\text{VaR}_{0.95} = S_0 - \text{Quantile}_{0.05}(S_T)$
- **Conditional VaR (CVaR / Expected Shortfall):** Average loss incurred in the worst 5% tail scenarios:
  $\text{CVaR}_{0.95} = \mathbb{E}[\, S_0 - S_T \mid S_T \leq \text{Quantile}_{0.05} \,]$

8. System Architecture & Mathematical Verification

Layered computational pipeline, zero-black-box design, and strict floating-point validation

Three-Tier Computational Architecture

1. **Mathematical Core (models/):** Pure functional implementations of TVM, amortizations, Newton-Raphson root-finders, and vectorized GBM simulations.
2. **Verification Layer (tests/):** Automated test suite enforcing domain boundaries and numerical convergence.
3. **Interactive Visual Interface (app.py):** Streamlit dashboard rendering dynamic probability distributions, cash flow schedules, and sensitivity sliders.

Design Principle: Zero Black Boxes

Every single algorithm in code maps directly to an axiomatic chapter in our theoretical monograph series.

Automated Testing & Benchmark Methodology

Academic research demands uncompromising reproducibility. Every module is verified against three criteria:

- **Independent Benchmark:** Deterministic models cross-checked against Microsoft Excel standards (PMT, IRR, PV) to $\varepsilon \leq \mathbf{10^{-7}}$.
- **Boundary Value Analysis:** Validating zero-interest limits ($r \to 0$), instantaneous horizons ($t \to 0$), and infinite compounding ($m \to \infty$).
- **Domain Guardrails:** Strict raising of ValueError on negative parameters, nonsensical loan tenors, and division by zero.

9. Academic Contributions & Research Milestones

Current progress, societal impact, and research roadmap for the Avishkar Convention

Completed Research Milestones

• **12-Volume Reference Monograph (56 Pages):** Fully authored, formatted, and compiled in theory_and_derivations/.
• **Analytical Verification:** Hand-solved academic case studies verifying all equations down to exact rupee arithmetic.
• **Open-Source Repository:** Publicly hosted on GitHub under MIT License with conventional semantic commits.

Societal Relevance & Consumer Protection

• **Democratizing Institutional Math:** Equipping common citizens with hedge-fund grade stochastic risk metrics.
• **Debt Trap Prevention:** Exposing predatory negative amortization structures before borrowers sign contracts.
• **Informed Wealth Planning:** Revealing actual probability densities of achieving financial independence.

Thank You! | Open for Questions & Mathematical Defense

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